

# Olist E-Commerce Business Insights Report

## 1. Project Objective

The goal of this analysis is to understand Olist's customers, orders, products, and sales performance using Python, Pandas, NumPy, and visualization libraries. The findings will help management make better decisions about inventory, marketing, and logistics.

## 2. Sales Analysis

- Top Categories by Quantity: Health & Beauty, Computers & Accessories, Bed & Bath Table dominate unit sales.
- Top Categories by Revenue: Electronics, Furniture & Decor, and Home Appliances generate the highest revenue.
- Top Products: Some products sell in large quantities but at low prices, while others generate high revenue despite fewer units.
- Monthly Trend: Sales peak during holiday/festival months, showing clear seasonality.

## 3. Customer Analysis

- Top States: São Paulo (SP), Rio de Janeiro (RJ), and Minas Gerais (MG) account for the majority of customers.
- Top Spenders: A small group of customers contributes disproportionately to revenue.
- Purchasing Behavior: Most customers place only one order, but a few are repeat buyers with multiple purchases.

## 4. Product & Category Findings

- Concentration: A handful of categories drive most of the business.
- Price Sensitivity: Higher-priced products tend to sell fewer units, showing weak/negative correlation between price and quantity.
- Opportunities: Bundling or discounting premium products could increase sales volume.

## 5. Delivery Findings

- Average Delivery Time: Around X days (from your calculation).
- Delays: A significant number of orders are delivered later than estimated.
- Impact: Late deliveries risk customer dissatisfaction and churn.

## 6. Business Insights (Q20)

Insight 1: Few categories drive most sales

- Evidence: Top 10 categories account for more than half of total revenue.
- Implication: Focus inventory and marketing on these categories.

Insight 2: Seasonal sales spikes

- Evidence: Monthly trend shows clear peaks in festive months.
- Implication: Plan campaigns and stock around seasonal demand.

Insight 3: Customer base concentrated in few states

- Evidence: SP, RJ, MG dominate customer distribution.
- Implication: Optimize logistics hubs in these states.

Insight 4: Delivery delays are common

- Evidence: Many orders delivered later than estimated.
- Implication: Improve courier partnerships and tracking.

Insight 5: Price vs. Quantity shows weak correlation

- Evidence: Scatter plot shows negative correlation.
- Implication: Offer bundles/discounts for expensive products.

## 7. Recommendations

1. Inventory Focus: Prioritize stocking top-selling categories → ensures consistent revenue.
2. Seasonal Strategy: Launch targeted campaigns during peak months → capture seasonal demand fully.
3. Regional Logistics: Strengthen delivery hubs in SP, RJ, MG → faster delivery, lower costs.
4. Delivery Improvement: Partner with reliable couriers and track delays → improves customer trust.
5. Pricing Strategy: Offer bundles/discounts for expensive products → increases sales volume.

## 8. Visualizations (minimum 6 required)

1. Order Status Distribution (bar chart)
2. Top Product Categories (bar chart)
3. Monthly Sales Trend (line chart)
4. Customer Distribution by State (bar chart)
5. Price vs Quantity (scatter plot)
6. Delivery Time Distribution (histogram)